UNIVERSITY OF TORONTO

# STA2112 Statistical Learning I — Homework 1 (v2)

Fall 2026: Lectures 1–4

Total: 100 points

Workload revision: October 1, 2026

Last name: _____ First name: _____

Student number: _____

## Scope and submission

Questions 1–5 combine conceptual reasoning and derivations (62 points). Question 6 is a ridge implementation and simulation exercise (13 points); Question 7 is a real-data comparison and short explanation (25 points). Lecture 5 material is not required. Use the deadline and submission location announced on Quercus.

- Submit a PDF writeup and one executed notebook with code and outputs. Use the supplied starter notebook and data; include package versions and seeds. The plot commentary in Question 7 is at most 150 words. Word, LaTeX, or neatly handwritten work is fine; LaTeX is not required. Hand-drawn or computer-drawn sketches are both acceptable.
- You do not need to reproduce the questions. Clearly label your answers by question number and part, and show your reasoning and intermediate steps. Questions 1–5 do not require code. For Question 6 implement ridge using basic linear algebra; for Question 7 use standard library estimators. Do not submit raw notebook output in place of explanations. In Question 4, complete both common parts and exactly two of the four SVM options; clearly identify your choices. Extra options carry no bonus marks.
- Homework is graded mostly for genuine completion and reasoning, in accordance with the syllabus. Point values indicate the relative weight of each part. Discussion is allowed, but writeups and code must be your own. Acknowledge resources. Generative AI may support conceptual learning, debugging, and writing feedback, but may not generate submitted answers, derivations, code, or the explanation.
- Additional ridge and SVM exercises are provided in *Homework 1 supplementary derivations*. These are optional, ungraded, and do not need to be submitted.

Use the terminology and results in Lectures 1–4, together with basic calculus, linear algebra, and probability. Any extra mathematical facts needed below are supplied in the question. Short, direct explanations are sufficient; you are not being asked to introduce additional theory. The optional supplement is not a prerequisite for this assignment. Hints and supplied facts are not additional tasks. Answer only the numbered requests; the displayed subpoint allocations sum to each part's total. You do not need to rewrite completed work to match the revised wording. Extra discussion requested only in an earlier version is no longer required.

| Question | Points | Main coverage |
|---|---|---|
| 1. Prediction targets and evaluation | 12 | Lectures 1–2 |
| 2. Regularization and shrinkage | 14 | Lecture 2 |
| 3. CV and the one-SE rule | 8 | Lectures 2 and 4 |
| 4. Classification; SVM options: choose two | 16 | Lecture 3 |
| 5. Trees and ensembles | 12 | Lecture 4 |
| 6. Implement and investigate ridge | 13 | Lecture 2 |
| 7. Compare models on Hitters | 25 | Lectures 1–4 |

## Question 1: Prediction targets and evaluation (12 points)

A hospital makes a prediction when a patient is admitted: will that patient be readmitted within 30 days after leaving the hospital? Its records contain several admissions per patient; each row is one admission. Candidate features include age at admission, measurements made at admission, patient ID, and total length of the current hospital stay. The model will be used for future admissions of patients not represented in the training data. Assume the records include admission and discharge dates and enough follow-up to determine these outcomes.

(a) **[4 points]**

(i) **[2 points]** Which proposed features would you exclude from the prediction model, and why?

(ii) **[2 points]** Why could a random split of admissions give a misleading evaluation for future admissions of previously unseen patients?

(b) **[4 points]** A valid development/test split has already been supplied: test admissions are later and patients do not overlap. All development outcomes were known by the training cutoff, which is the date when the model would be trained.

For example, if a patient was discharged only 10 days before that date and has not yet been readmitted, the outcome cannot be labeled 0: readmission could still occur during the remaining 20 days. Outcomes learned after the cutoff cannot be used, even if they are available when looking back at the records later.

You do not need to design this split. For each practice below, briefly state the problem and its correction.

(i) **[2 points]** Before CV, the analyst uses all development and test rows to fit standardization and select predictors using their association with the outcome.

(ii) **[2 points]** The analyst chooses $k$ by CV and reports the smallest CV error as the final evaluation, without using the independent test set.

(c) **[4 points]** At a fixed input $x$, let $Y \sim \text{Bernoulli}(p(x))$ be a fresh outcome, independent of training data $D$. Put $\mu(x) = E_D[\widehat{p}_D(x)]$, the average predicted probability over new training samples of the same size. The input $x$ and the distribution are fixed; $D$ and the independent new outcome $Y$ are random.

(i) **[3 points]** Derive

$$E_{D,Y}[(Y - \widehat{p}_D(x))^2] = p(x)(1 - p(x)) + (\mu(x) - p(x))^2 + \text{Var}_D(\widehat{p}_D(x)).$$

(ii) **[1 points]** If $0 < p(x) < 1$, can an exact fit to the training outcomes eliminate the randomness of a new outcome? Give one sentence using the decomposition.

## Question 2: Regularization and shrinkage (14 points)

Each column of $X$ has mean zero; the intercept is fixed at zero. Let $L(\beta) = \|y - X\beta\|_2^2$ and let $P(\beta)$ be either $\|\beta\|_2^2 = \sum_j \beta_j^2$ (ridge) or $\|\beta\|_1 = \sum_j |\beta_j|$ (lasso). Assume all minimizers below exist.

(a) **[4 points]** Let $\widehat{\beta}(\lambda)$ minimize $L(\beta) + \lambda P(\beta)$, where $\lambda > 0$.

(i) **[2 points]** Prove that it also minimizes $L(\beta)$ subject to $P(\beta) \le P(\widehat{\beta}(\lambda))$.

(ii) **[2 points]** Conversely, let $\widetilde{\beta}$ be any minimizer of this same constrained problem. Prove that it also minimizes $L(\beta) + \lambda P(\beta)$ for the same $\lambda$.

(b) **[3 points]** For $0 < \lambda_1 < \lambda_2$, prove $P(\widehat{\beta}(\lambda_2)) \le P(\widehat{\beta}(\lambda_1))$. In other words, increasing the penalty weight cannot increase the total penalty of the fitted coefficient vector.

(c) **[4 points]** The ridge formula is supplied: $\widehat{\beta}^{\text{ridge}}(\lambda) = (X^\top X + \lambda I)^{-1} X^\top y$. For one centered predictor with $\sum_i x_i^2 = s > 0$ and $y_i = x_i\beta + \varepsilon_i$, where independent errors have mean zero and variance $\sigma^2$, this becomes $\widehat{\beta}^{\text{ridge}}(\lambda) = \sum_i x_i y_i/(s + \lambda)$. Keep $X$ fixed.

(i) **[2 points]** Derive $E[\widehat{\beta}^{\text{ridge}}(\lambda) \mid X] - \beta$, its bias.

(ii) **[2 points]** Derive $\text{Var}(\widehat{\beta}^{\text{ridge}}(\lambda) \mid X)$.

(d) **[3 points]** Interpret the formulas in part (c), keeping $s$, $\beta$, and $\sigma^2$ fixed. No derivatives are required.

(i) **[2 points]** As $\lambda \ge 0$ increases or decreases, how do the *absolute bias* and the variance change? Briefly justify your answer using the formulas in (c).

(ii) **[1 points]** Explain how $\lambda$ controls the bias and variance tradeoff. Does a smaller variance alone guarantee a smaller mean squared error for estimating $\beta$?

## Question 3: CV and the one-standard-error rule (8 points)

The following scores were computed using CV within the training/development data, not the final test set. The reported SE is the fold standard deviation divided by the square root of the number of folds.

| | Candidate | CV MSE | Rough SE |
|---|---|---|---|
| $k$NN | $k = 3$ | 0.20 | 0.02 |
| | $k = 9$ | 0.18 | 0.03 |
| | $k = 25$ | 0.20 | 0.02 |
| Regression tree | 3 leaves | 0.43 | 0.02 |
| | 9 leaves | 0.41 | 0.03 |
| | 20 leaves | 0.44 | 0.04 |

The two blocks concern different prediction tasks; do not compare their MSEs. Larger $k$ means a smoother $k$NN rule; fewer leaves means a simpler tree.

(a) **[4 points]** Apply the one-SE rule. In each answer, report the minimum-CV choice, the cutoff (minimum CV MSE plus its SE), and the simplest eligible choice.

(i) **[2 points]** The $k$NN block.

(ii) **[2 points]** The regression-tree block.

(b) **[4 points]** Give a brief explanation for each answer; no formal test is required.

(i) **[2 points]** Does being within the one-SE cutoff prove equal expected prediction errors for the eligible candidates?

(ii) **[2 points]** For a fixed fitted model, must two independent test samples of the same size from the same population give identical test MSEs?

## Question 4: Classification, margins, and decisions (16 points)

Use $y \in \{0, 1\}$ for logistic regression, but $y_i \in \{-1, +1\}$ for SVM. Complete both common parts (6 points), then exactly two SVM options (10 points). All parts use Lecture 3. The SVM options below give the steps or formulas needed; no additional constrained-optimization theory is required.

### Common parts

(a) **[3 points] From log-loss derivatives to fitting logistic regression.** For training observations $(x_i, y_i)$, logistic regression uses scores $z_i = x_i^\top \beta$ and probabilities $p_i = (1 + e^{-z_i})^{-1}$; include a constant 1 in $x_i$ for the intercept. Fitting the model means choosing the shared coefficient vector $\beta$ to minimize

$$J(\beta) = \sum_{i=1}^{n} \ell(y_i, x_i^\top \beta), \qquad \ell(y, z) = \log(1 + e^z) - yz.$$

The two calculations below provide the ingredients for updating $\beta$. For one observation, write $p = (1 + e^{-z})^{-1}$.

(i) **[2 points]** Differentiate once with respect to $z$, expressing the result using $p$.

(ii) **[1 points]** Differentiate again, expressing the result using $p$.

**How these calculations are used (provided explanation).** At the current $\beta$, the chain rule combines your two scalar derivatives into

$$g(\beta) = \nabla J(\beta) = \sum_{i=1}^{n} \frac{\partial \ell(y_i, z_i)}{\partial z_i} x_i, \qquad H(\beta) = \nabla^2 J(\beta) = \sum_{i=1}^{n} \frac{\partial^2 \ell(y_i, z_i)}{\partial z_i^2} x_i x_i^\top.$$

The *gradient* $g$ is the vector of first derivatives with respect to the coefficients; the *Hessian* $H$ is the matrix of second derivatives, describing curvature.

**Gradient descent** updates $\beta_{\text{new}} = \beta - \eta g(\beta)$. It moves opposite the gradient, the direction of steepest increase in loss. The positive step size $\eta$ is chosen small enough to decrease the loss.

**Newton's method** also uses curvature: when $H$ is positive definite, it minimizes a local quadratic approximation to $J$, giving $\beta_{\text{new}} = \beta - H(\beta)^{-1} g(\beta)$. For a single parameter, this divides the slope by the curvature. The step can be shortened if the full step does not decrease $J$.

After either update, recompute the scores and probabilities and repeat. Thus your derivatives are used to fit the model, not just to describe the loss. These methods update the shared coefficients, not each probability independently. The formulas above are supplied context; no further derivation or code is required.

(b) **[3 points]** A false negative costs 4 and a false positive costs 1; correct decisions cost zero. Assuming $p(x)$ is the true conditional probability:

(i) **[2 points]** Derive the optimal decision threshold by comparing expected costs.

(ii) **[1 points]** At $p = 0.30$, give the decisions under this rule and threshold $0.50$.

## Question 4 continued: SVM options (choose two, 5 points each)

Each option is self-contained. The distance formula may be used without proof in option B, even if you do not choose option A. The explanations introduce the ideas behind the calculations; answer only the numbered requests.

A. **Distance from a point to the decision boundary.** Let $H = \{u : w^\top u + b = 0\}$ with $w \neq 0$, and fix a point $x$. We want to turn the score $w^\top x + b$ into a geometric distance.

**Geometric background.** A *normal vector* is perpendicular to every direction within the boundary. The Pythagorean identity says that if $v^\top r = 0$, then

$$\|v + r\|^2 = \|v\|^2 + \|r\|^2,$$

because the cross term $2v^\top r$ is zero. You may use this identity. The first part explains why the shortest connection to the boundary is perpendicular; the second turns that geometry into a distance formula.

(i) **[2 points]** Show that $w$ is a normal vector to $H$. Then choose $c$ so that $z_* = x - cw \in H$ and explain why $z_*$ is the closest point in $H$ to $x$. Calculate $c$ only in part (ii). *Hint:* first subtract the boundary equations for two points in $H$. Then, for any $u \in H$, write $x - u = (x - z_*) + (z_* - u)$ and use the Pythagorean identity to compare distances. When $x \in H$, the distance is zero.

(ii) **[3 points]** Substitute $z_* = x - cw$ into $w^\top z_* + b = 0$ and solve for $c$. Then calculate $\|x - z_*\| = \|cw\|$ to obtain the distance formula below, explaining why the absolute value is needed:

$$\operatorname{dist}(x, H) = \|x - z_*\| = \frac{|w^\top x + b|}{\|w\|}.$$

B. **Maximum margin.** Assume both labels occur and some $(w, b)$ has all signed scores $y_i(w^\top x_i + b) > 0$. Let $m = \min_i y_i(w^\top x_i + b) > 0$. The geometric margin is the distance from the decision boundary to the nearest training point. A separator is normalized when its smallest signed score is 1.

**Why normalize?** Multiplying both $w$ and $b$ by a positive constant changes all scores but not their signs or the boundary. Thus a large raw score alone does not mean a point is far from the boundary. Normalization removes this arbitrary scale so that we can compare different separators. The distance formula, supplied here independently of option A, is $\mathrm{dist}(x_i, H) = |w^\top x_i + b|/\|w\|$. For a correctly classified point, the score has the same sign as $y_i$, so $|w^\top x_i + b| = y_i(w^\top x_i + b)$. Therefore

$$\text{geometric margin} = \min_i \frac{y_i(w^\top x_i + b)}{\|w\|} = \frac{m}{\|w\|}.$$

When $m = 1$, this becomes $1/\|w\|$. The goal below is to connect this geometric quantity to the objective and constraints used to fit an SVM.

(i) **[2 points]** Set $\widetilde{w} = w/m$ and $\widetilde{b} = b/m$. Show that the equation for the boundary is unchanged and that $\min_i y_i(\widetilde{w}^\top x_i + \widetilde{b}) = 1$.

(ii) **[3 points]** Using the normalized margin $1/\|w\|$, explain why choosing a separator with the largest geometric margin is equivalent to solving

$$\min_{w,b} \tfrac{1}{2}\|w\|^2 \quad \text{subject to} \quad y_i(w^\top x_i + b) \geq 1 \text{ for all } i.$$

*Hint:* if all signed scores exceed 1, divide both $w, b$ by their smallest score; what happens to the objective? No separate convexity proof is required.

## Question 4 continued: SVM options C and D

Choose exactly two options in total from A, B, C, and D.

C. **Slack and hinge loss.** For $C > 0$, minimize $\|w\|^2/2 + C\sum_i \xi_i$ subject to $\xi_i \geq 0$ and $y_i(w^\top x_i + b) \geq 1 - \xi_i$.

**What does slack mean?** The signed score $y_i(w^\top x_i + b)$ is positive when the point is correctly classified; reaching 1 also meets the normalized margin requirement. A nonnegative slack $\xi_i$ relaxes that requirement for observation $i$. A point can be on the correct side but inside the margin, so positive slack need not mean misclassification. The term $C\sum_i \xi_i$ makes these relaxations costly, with larger $C$ putting more weight on margin violations relative to $\|w\|^2/2$. For fixed $w, b$, the geometry is fixed. The calculation below removes unnecessary slack and expresses its smallest cost directly as a loss on the signed scores.

(i) **[2 points]** Hold $w, b$ fixed, so all signed scores are fixed. Find the smallest feasible value of each $\xi_i$ and explain why it minimizes the objective over the slack variables.

(ii) **[3 points]** Substitute those values into the objective to obtain a formula involving only $w, b$. Explain why minimizing this formula gives the same optimal choices of $w, b$ as the problem with slack variables.

D. **Support vectors and kernels.** Use the fitted weight representation given in Lecture 3, without proving how the weights are obtained:

$$w = \sum_{i=1}^{n} a_i y_i x_i, \qquad 0 \le a_i \le C.$$

The points with $a_i > 0$ are called support vectors. Here the $a_i$ are fitted weights, not probabilities. A zero weight means that observation contributes no term to this representation of $w$. The classifier predicts the sign of $f(x)$.

**Why replace inner products?** A linear classifier in transformed features $\phi(x)$ uses $\phi(x_i)^\top \phi(x)$ wherever the original linear classifier uses $x_i^\top x$. A kernel computes this transformed inner product directly: $K(x_i, x) = \phi(x_i)^\top \phi(x)$. It is therefore a way to change the feature representation without explicitly constructing all its coordinates. The supplied RBF kernel is one valid choice; it gives a similarity that decreases with distance. Refitting in this representation can produce a nonlinear boundary in the original inputs. The tasks below show how the prediction formula changes.

(i) **[2 points]** Substitute the sum for $w$ into $f(x) = w^\top x + b$. Expand the transpose of the sum to express the score using the inner products $x_i^\top x$.

(ii) **[3 points]** Replace the inner products by $K(x_i, x) = \exp\{-\gamma \|x_i - x\|^2\}$, where $\gamma > 0$, to write the RBF score. Explain in one sentence how larger $\gamma$ changes how quickly a kernel contribution decreases as $x$ moves away from $x_i$. The weights and intercept are refitted for the new kernel; you do not need to derive them.

If you already completed the earlier derivation-based option D, it remains an acceptable option D answer; you do not need to redo it.

## Question 5: Trees, averaging, and regularized corrections (12 points)

(a) **[2 points]** A binary-outcome leaf has observed positive proportion $\widehat{p}$. The labels are 0 or 1, and every observation in the leaf receives the same probability prediction $\widehat{p}$, without rounding it to a class label. If the leaf has $n$ observations, training MSE means $n^{-1}\sum_{i \text{ in leaf}}(y_i - \widehat{p})^2$. Grouping this average into the positive and negative observations connects the squared-error criterion to how mixed the leaf is. Show that predicting $\widehat{p}$ gives training MSE $\widehat{p}(1-\widehat{p})$.

(b) **[4 points]** At a fixed $x$, suppose each of $B$ tree predictions has variance $\sigma^2$ and every distinct pair has correlation $\rho$. Here variance describes how a fitted prediction would change across repeated training datasets and tree randomization, not the variation of outcomes across different inputs. The trees predict the same $x$ and can move together because they use overlapping information. This simplified common-variance, common-correlation model lets us isolate what averaging can and cannot remove; $\sigma^2$ here is tree-prediction variance.

(i) **[3 points]** Derive

$$\operatorname{Var}\left(\frac{1}{B}\sum_{b=1}^{B}\widehat{f}_b(x)\right) = \sigma^2\left[\rho + \frac{1-\rho}{B}\right].$$

*Hint:* expand the variance of the sum into $B$ variance terms and $B(B-1)$ covariance terms, using covariance $\rho\sigma^2$ for each distinct pair.

(ii) **[1 points]** Give its limit as $B \to \infty$ for fixed $\rho$.

(c) **[6 points]** At boosting step $t$, the current prediction is $F_{t-1}(x_i)$ and its residual is $r_i = y_i - F_{t-1}(x_i)$. Fix one leaf: $I_j$ is the set of observations in it and $n_j$ is their number. Minimize

$$\frac{1}{2}\sum_{i\in I_j}(r_i - w)^2 + \frac{\lambda}{2}w^2, \qquad \lambda \geq 0.$$

The scalar $w$ is a shared *correction* for this leaf, not a replacement for the current prediction. Before applying a learning rate, adding $w$ would change each residual from $r_i$ to $r_i - w$, which explains the first term. The second term discourages a large correction. Write $\bar{r}_j = n_j^{-1}\sum_{i\in I_j} r_i$. Because the coefficient of $w^2$ is $(n_j + \lambda)/2 > 0$, the quadratic opens upward and its zero derivative gives the unique minimum. After finding that correction $w_j$, the learning rate $\nu$ controls how much to use: the new prediction in the leaf is $F_t(x_i) = F_{t-1}(x_i) + \nu w_j$.

(i) **[2 points]** Differentiate to find the minimizing correction $w_j$.

(ii) **[4 points]** For $\bar{r}_j = 0.80$, $\lambda = 5$, and $\nu = 0.10$, calculate the amounts $\nu w_j$ added for leaf sizes 5 and 50. Explain why the smaller leaf receives a smaller update despite having the same mean residual, and how this differs from reducing the learning rate.

## Question 6: Implement ridge and study prediction error (13 points)

Use the supplied fixed, column-centered design matrix $X$ with 60 rows and 5 columns. Its first two columns are strongly correlated. For this controlled simulation, the intercept is known to be zero and $\beta = (1, 1, 0, 0, 0)^\top$. Generate $R = 500$ independent responses $y^{(r)} = X\beta + \varepsilon^{(r)}$ with $\varepsilon^{(r)} \sim N(0, I_{60})$, using seed 2112. Keep $X$ fixed and use the same responses for all $\lambda \in \{0, 2, 20, 200\}$. Make predictions at $x_0 = (1, 0, 0, 0, 0)^\top$, with true mean $f_0 = x_0^\top \beta = 1$. Use the same objective as Question 2: $\|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2$. The starter generates these repeated datasets. Its existing `Y0` variable may be ignored: a separate noisy-outcome experiment is not required. This is the repeated-training-sample bias–variance experiment from Lecture 2, now using ridge instead of $k$NN.

**What this experiment measures.** We know $\beta$ only because we generate the data; the fitted model still estimates it from each noisy response vector. Holding $X$ and $x_0$ fixed isolates the effect of new training noise. Reusing the same responses across penalties gives a direct comparison on the same datasets. The target is the fixed mean $f_0$, not a new noisy outcome, so the reported MSE does not include new test-outcome noise.

(a) **[5 points]**

(i) **[3 points]** Implement the supplied ridge formula from Question 2 using a linear system solve, not a library ridge fit or explicit matrix inverse. You may use `numpy.linalg.solve`; do not estimate an intercept or recenter responses. This computes the vector that satisfies the ridge linear equations directly; there is no need to form a matrix inverse.

(ii) **[2 points]** For the first simulated response, report the largest absolute entry of the following vector for each $\lambda$, checking it is below $10^{-8}$:

$$(X^\top X + \lambda I)\widehat{\beta} - X^\top y.$$

This is a numerical check that your coefficients solve the intended equations, not a prediction-error score. A small value does not by itself mean that the fitted model predicts well.

(b) **[6 points]** For each $\lambda$, estimate the bias, variance, and MSE of $\widehat{f}^{(r)}(x_0) = x_0^\top \widehat{\beta}^{(r)}(\lambda)$ across repetitions. Let $\bar{f} = R^{-1} \sum_r \widehat{f}^{(r)}(x_0)$. Use

$$\widehat{\text{Bias}} = \bar{f} - f_0, \qquad \widehat{\text{Var}} = \frac{1}{R}\sum_r (\widehat{f}^{(r)}(x_0) - \bar{f})^2, \qquad \widehat{\text{MSE}}_f = \frac{1}{R}\sum_r (\widehat{f}^{(r)}(x_0) - f_0)^2.$$

The variance uses divisor $R$, not $R - 1$. Here $\bar{f}$ is the center of the predictions over simulated training datasets. Bias measures its displacement from $f_0$; variance measures the predictions' spread around $\bar{f}$; MSE measures their squared distance from $f_0$. These are averages over the $R$ simulated fits, which is why the displayed variance uses divisor $R$.

(i) **[4 points]** Report one row per $\lambda$: bias, squared bias, variance, and $\widehat{\text{MSE}}_f$.

(ii) **[2 points]** Plot squared bias, variance, and $\widehat{\text{MSE}}_f$ against $\lambda$.

(c) **[2 points]** Use the results from (b) to briefly describe the observed bias–variance tradeoff.

## Question 7: A reproducible Hitters comparison (25 points)

Use the provided Hitters data (263 players with observed salary; one row per player). Predict *log salary* from these same eight inputs for every method: Years, Hits, AtBat, HmRun, Runs, RBI, Walks, PutOuts. Use the supplied row assignments: 210 development players (used to fit and select models), 53 test players (used only for final evaluation), and five fixed development folds. The row identifier identifies the player; it is not a predictor. Assume the eight inputs are available when predicting. No additional feature engineering or data is required. Keep test outcomes out of fitting, preprocessing, and selection. Use seed 2112; break CV ties by the listed order. Report package versions; small numerical differences are acceptable.

**Why separate development and test data?** CV uses the development players to compare settings and choose a predictor. Because those scores influence the choice, the untouched test players provide a separate final evaluation. Keeping the folds identical across methods makes the comparison use the same training and validation players.

(a) **[4 points]**

(i) **[2 points]** Load the supplied assignments and report the development/test counts and the five validation-fold counts.

(ii) **[2 points]** Does this split establish performance in a future baseball season? Explain briefly.

(b) **[9 points]**

(i) **[6 points]** Use the same five folds to compute CV MSE on log salary for the candidates below and a training-fold-mean baseline. Fit ridge scaling within each training fold; trees need no scaling. For each candidate, hold out one of the five folds, fit on the other four, predict the held-out players, and compute their mean squared error. Repeat for all five folds and average the five MSEs. For ridge, estimate feature means and standard deviations only from the four training folds and apply those same values to the held-out fold. These preprocessing quantities are part of the fitted procedure. The baseline likewise predicts the mean log salary of the current training folds, not a mean computed using validation or test outcomes.

(ii) **[3 points]** Record each family's CV-selected setting and your overall CV-selected predictor (including the baseline as a candidate) *before* computing test scores. A compact table is sufficient; no separate essay is required.

| Family | Required candidates and fixed settings |
|---|---|
| Ridge | Standardize using the current training fold, then fit slopes and an unpenalized intercept $b_0$. Objective: $\sum_i (y_i - b_0 - x_i^\top \beta)^2 + \alpha \lVert\beta\rVert_2^2$; $\alpha \in \{0.1, 1, 10, 100, 1000\}$. With the same data and preprocessing, $\alpha = \lambda$ matches Question 2. |
| Single tree | Maximum leaves $\in \{3, 9, 20\}$; minimum leaf size 5. |
| RF / bagging | 200 bootstrap trees; minimum leaf size 5; no depth cap; max features per split $\in \{1, 3, 8\}$. The value 8 is bagging with these settings. |
| Boosting | Squared loss; at most 3 leaves per tree; minimum leaf size 5; $\eta$ is the learning rate and $M$ the number of trees; $(\eta, M) \in \{(0.03, 100), (0.03, 300), (0.10, 100), (0.10, 300)\}$. |

(c) **[5 points]**

(i) **[2 points]** Refit each CV-selected family on all development observations. Once all choices are frozen, evaluate the four selected models and the mean baseline (now using the full development mean) on the test set in one final pass. CV fits are temporary models for comparing settings. This refit trains each selected setting anew using all available development observations; it does not average the five CV models.

(ii) **[3 points]** Add test MSEs to the table of settings and CV MSEs from (b). Do not switch your chosen predictor after seeing test scores.

(d) **[7 points]** Use the predictor chosen *before* testing.

(i) **[4 points]** Plot test residuals (observed minus predicted log salary) against predicted log salary, including a horizontal line at zero. A positive residual means the model underpredicted log salary; a negative one means it overpredicted. The plot checks where predictions miss, rather than only how large the average squared error is. Use it for the requested description, not to retune the model on the test set.

(ii) **[3 points]** Describe one visible pattern, or the absence of a clear pattern, and what it suggests about the predictions. Two to four sentences are sufficient (at most 150 words); no broader model-comparison essay is required.